

How the scan flipflops were selected for partial scan?

- 1. SCOAP ranking:** I synthesized b14.vhd in full-scan mode and ran `report_primitives -all` in TMAX to dump CC0, CC1, CO for every D-FF into a file. I combined them as $D=CC0+CC1+CO$ and produced an ordered list from “easiest” to “hardest.” (lowest to highest SCOAP values)
- 2. Testing both ends of the list:** Instead of blindly choosing the top-K hardest flops, I built candidate K-sets from both the lowest- and highest-SCOAP regions to see which side actually helped the overall metric M in terms of differing area, test coverage or pattern length.
- 3. Keep the coverage-critical flops:** Continuous removal tests showed that excluding `state_reg` or `IR_reg[23]` collapsed test coverage to ~30 %. Those flops were therefore locked in the scan chain, independent of SCOAP rank.
- 4. Register balancing:** Putting all the flip flop bits of one large register into or completely out of the scan chain affected test coverage and the number of patterns in some instances. So, instead, in practice I chose K-sets to sample flops from several different registers.
- 5. Final selection:** The best-M configuration kept a single scan chain and **removed** via `set_scan_element false`, the **20 flip-flops with the highest SCOAP difficulty scores**. Leaving the easier flops in scan cut the pattern count sharply once `set_atpg -coverage 80` was applied.

How design exploration was conducted?

I iterated through a loop of edit → synthesize → ATPG → measure MMM until the metric peaked.

- 1. Edit step:** For each trial value of K (10, 20, 40, 80, ...), I commented flops out of the scan chain with `set_scan_element false { ... }`, following the selection rules as described above.
- 2. Synthesis step: `synthesis_pscan.tcl`** regenerated the netlist (b14_pscan.vg) with one scan chain. From the reports I captured new area (A) and max-chain length (L). `compile_ultra -scan` was used as it generated really low area in comparison to `compile -scan`
- 3. ATPG step:** Running `tmax_pscan.tcl` produced test coverage (TC) and pattern count (N). I began with no coverage goal, then tried `set_atpg -coverage` values of 99, 90, 85, 80, etc. to observe which coverage value gave the best trade-off between high test coverage and low pattern count.
- 4. Metric step:** For each run I plugged TC, A, N, L (with $P=3$, since scan chain was 1) into the formula provided to calculate M.
- 5. Key findings.**
 - a. One scan chain always beat two; a second chain raised other metrics like P more than it lowered N.

- b. Normally excluding the lowest-SCOAP flops from the scan chain resulted in high M values due to slightly higher TC, slightly lower area and pattern count in some cases .
- c. Certain control flops (wr_reg, rd_reg) inflated pattern count when left in scan, so they were purposefully excluded.
- d. Certain flops like state_reg or IR_reg[23] were key to test coverage as mentioned in the above answer
- e. The pivotal difference maker was set_atpg -coverage: (On an example where the same 20 lower SCOAP flops were excluded from scan)
 - No goal → TC ≈ 91%, N = 241 → M≈332
 - Set coverage 85 → TC ≈ 86 %, N = 64 → M≈1180M

Following this cycle, the run where K = 20 highest SCOAP flip flops were excluded from scan chain and set_atpg -coverage 80 delivered the best overall M while still keeping TC well above the required 40 %.

Fill the table below which includes the design points that were explored by you to find your best design, highlighting the best column (highest M)

	Partial 1 (K=10) Low SCOAP ffs excluded from scan No set coverage	Partial 2 (K=20) High SCOAP ffs excluded from scan No compile_ultra -scan and set coverage used	Partial 3 (K=20) Low SCOAP ffs excluded from scan No set coverage	Partial 4 (K=20) Low SCOAP ffs excluded from scan Using set coverage 85	Partial 5 (K=20) High SCOAP ffs excluded from scan Using set coverage 85	Partial 6 (K=20) Low SCOAP ffs excluded from scan Excluding wr_reg rd_reg in scan chain Using set coverage 85	Partial 7 (K=20) High SCOAP ffs excluded from scan Using set coverage 75	Partial 8 (K=30) Low SCOAP ffs excluded from scan No set coverage
TC	95.57	89.32	91.1	85.93	86.68	86.62	75.1	85.98
A	7589.668	11863.187	8565.763992	8565.763992	8564.96247	8565.605664	8564.96247	7518.081823
N	254	257	241	64	64	96	32	178
L	205	195	195	195	195	195	195	185
P	3	3	3	3	3	3	3	3
#Chains	1	1	1	1	1	1	1	1
@Scan FFs	205	195	205	195	195	195	195	185
M	355.2117933	220.6763287	332.4120551	1180.701918	1191.118581	793.4698054	2063.98259	510.1228868